

The Second-Order Tangent Cone of Compact Weyl–Maxwell Gravity

Moment Maps, Secular Corrections, and Bounded Resonances

GPT-5.6.sol*

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Abstract

Linearized solutions of a nonlinear gauge theory need not be tangent to exact solutions. We classify this failure at second order for finite harmonic configurations of Weyl–Maxwell gravity on the compact magnetically supported product $\mathbb{R}_t \times S^1 \times S^2$. The certified linear inventory contains the ordinary Einstein–Maxwell branches, two additional fourth-order branches in each generic parity, exceptional dipoles, a homogeneous generalized block, and axial twist modes.

For corrections that are smooth, spatially periodic, and finite exponential-polynomials in time, the complete second-order obstruction space is five-dimensional. Its covectors are precisely the Taub pairings with time translation, circle translation, and the three sphere rotations. Thus the complete finite-support smooth tangent cone is

$$\mathcal{Z}_2^{\text{smooth}} = \{u : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.$$

This is a necessary-and-sufficient theorem in the declared correction class, not merely a list of examples.

Bounded corrections obey additional conditions. Polynomial-growth source coefficients and nonzero-shell adjoint pairings survive independently of the five moment maps. We classify the complete standard generalized-zero subcone, prove that electric-charge and flat-Wilson oscillator columns are bounded-removable, and determine the full circumference column: a nonzero circle-radius tangent is compatible with a finite oscillator precisely when that oscillator has zero circle momentum. A repaired full-time circumference-velocity fixture also exposes a nonzero polar polynomial obstruction that was invisible at $t = 0$. The remaining bounded oscillator columns are stated as open gates. Consequently smooth second-order extendibility is a moment-map theorem, whereas bounded extendibility is a strict resonance problem.

All results are LOCAL-ALGEBRAIC and REDUCED-MODE. They do not establish an all-orders family, a causal retarded correction, a final residual quotient, or a quantum state space.

AI authorship and computational accountability

The named AI systems developed the programme, derivations, symbolic implementations, independent verification architecture, claim audits, and this manuscript. Asger Alstrup Palm commissioned and coordinated the work and is the corresponding human contact. Every theorem stated as certified below is bound to an exact machine-readable certificate and a separately implemented verifier. Exact rational or algebraic arithmetic is used for all ranks, cokernels, polynomial reductions, and source remainders. The manuscript remains conditional on final human review of its mathematical scope, references, and exposition.

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Contents

1	Question, answer, and claim boundary	2
1.1	Dependency typing	3
2	Theory and common background	3
3	The complete finite harmonic carrier	3
4	Smooth tangent-cone theorem	4
4.1	Proof architecture	4
5	What changes in the bounded category	5
6	Natural transport columns	5
7	Exceptional resonance as an independent obstruction	6
8	Current bounded frontier	6
9	Interpretation	7
10	What is not proved	8
11	Certificate and reproduction ledger	8
12	Outlook	9

1 Question, answer, and claim boundary

Let $\Phi(s)$ be a putative curve of solutions of the Weyl–Maxwell equations $\mathcal{E}(\Phi) = 0$, expanded as

$$\Phi(s) = \bar{\Phi} + su + \frac{s^2}{2}v + O(s^3).$$

The first two equations are

$$\mathcal{L}u = 0, \quad \mathcal{L}v = -\mathcal{E}_{\bar{\Phi}}^{(2)}(u, u), \quad \mathcal{L} := D\mathcal{E}_{\bar{\Phi}}. \quad (1)$$

The first equation describes the linear solution space. The second asks whether the quadratic source lies in the image of the linearized operator in a specified function class. Changing that function class changes the answer: a resonant source may have a smooth secular primitive but no bounded primitive.

The paper answers the following finite-harmonic question.

For the complete certified linear inventory on the compact product, which real finite-support tangents admit a spatially periodic second-order correction, and which additional conditions appear when that correction is required to remain bounded in time?

The answer has two layers:

$$u \in \mathcal{Z}_2^{\text{smooth}} \iff \mu_H = \mu_{P_x} = \mu_{J_1} = \mu_{J_2} = \mu_{J_3} = 0, \quad (2)$$

$$u \in \mathcal{Z}_2^{\text{bounded}} \implies u \in \mathcal{Z}_2^{\text{smooth}}, \quad \mathcal{P}_{j,r}(u) = 0, \quad \mathcal{R}_{j,a}(u) = 0. \quad (3)$$

Here $\mathcal{P}_{j,r}$ are positive polynomial-growth coefficients and $\mathcal{R}_{j,a}$ are reduced adjoint pairings on nonzero characteristic shells. The first line is complete in the declared carrier. The paper gives several complete columns of the second line but does not claim the complete common zero locus of all bounded obstructions.

1.1 Dependency typing

The spacetime is Lorentzian, but the proof is an exact harmonic and finite-module calculation. It carries the dependency tags

$$\text{LOCAL-ALGEBRAIC} \quad \text{REDUCED-MODE}.$$

No theorem below carries LORENTZIAN-CAUSAL. In particular, a finite secular right inverse is not a retarded Green operator, and a bounded quasiperiodic correction is not an all-orders nonlinear solution.

2 Theory and common background

We use the compactified Plebański–Hacyan product

$$M = \mathbb{R}_t \times S_L^1 \times S^2, \quad (4)$$

with unit sphere curvature and a parallel unit magnetic flux on a fixed compact $U(1)$ bundle. With the conventions of the companion linear phase-space paper, the background solves both Einstein–Maxwell and Weyl–Maxwell theory. The latter has local gauge group

$$\text{Diff}(M) \times \text{Weyl}(M) \times U(1).$$

The spatial slice $\Sigma = S^1 \times S^2$ is compact. Its background stabilizer algebra contains

$$H = \partial_t, qquad P_x = \partial_x, qquad J_1, J_2, J_3 \in \mathfrak{so}(3).$$

For a linear tangent u , the corresponding covariant phase-space moment maps are quadratic forms

$$\mu_X(u) = \frac{1}{2} \Omega(u, \mathcal{L}_X u), \quad X \in \{H, P_x, J_1, J_2, J_3\}.$$

On a closed slice, the covariant Taub bridge identifies these with the pairings of the quadratic field-equation source against the stabilizer cokernel covectors.

3 The complete finite harmonic carrier

The real finite-support carrier V_{fin} is the direct sum of the following certified linear blocks:

1. generic Einstein and extra primaries for every $\ell \geq 2$, parity, magnetic label, and compact momentum;

2. standard and extra exceptional dipoles for every compact momentum;
3. the axial twist Jordan block;
4. the six-coordinate homogeneous generalized block; and
5. the electric and Wilson global directions on the fixed bundle.

Reality pairs opposite frequencies and momenta. A finite input therefore produces a finite quadratic source. Its complete output decomposition is indexed by

$$(L, M, K, \Omega, \text{parity}).$$

Angular Clebsch–Gordan selection gives finitely many L , while temporal and circle products give finite sums of input frequencies and momenta.

Definition 3.1 (Correction classes). Let $\mathcal{X}_{\text{smooth}}$ be the spatially periodic fields whose time coefficients are finite sums of $t^p e^{-i\Omega t}$. Let $\mathcal{X}_{\text{bounded}} \subset \mathcal{X}_{\text{smooth}}$ be the subclass with $p = 0$. Define

$$\mathcal{Z}_2^{\mathcal{X}} = \{u \in V_{\text{fin}} : \exists v \in \mathcal{X}, \mathcal{L}v = -\mathcal{E}^{(2)}(u, u)\}.$$

The distinction is physical as well as analytic. A secular coefficient is smooth for every finite time but prevents a uniformly bounded perturbative expansion. It may signal a nearby family with shifted frequency, a required renormalization of parameters, or genuine resonant growth. The certificate records only the exact second-order disposition.

4 Smooth tangent-cone theorem

Theorem 4.1 (Complete finite-support smooth tangent cone). *For the complete carrier V_{fin} above,*

$$\boxed{\mathcal{Z}_2^{\text{smooth}} = \{u \in V_{\text{fin}} : \mu_H(u) = \mu_{P_x}(u) = \mu_{J_1}(u) = \mu_{J_2}(u) = \mu_{J_3}(u) = 0\}.} \quad (5)$$

Every point of this cone admits a spatially periodic second-order correction whose time dependence is a finite exponential-polynomial. Conversely, each of the five moment maps must vanish for such a correction to exist.

4.1 Proof architecture

The proof is blockwise but exhaustive.

Nonzero output blocks. Direct exceptional Fourier complexes and generic Smith modules remove every off-shell output algebraically. On a characteristic shell, the scalar factor admits a finite secular primitive. Thus resonance does not create a new cokernel in $\mathcal{X}_{\text{smooth}}$.

Homogeneous zero block. The only potentially non-surjective rows reduce to ordinary polynomial operators. The operators D_t^4 and D_t^2 are surjective on the finite source polynomials generated at quadratic order, whose degree is at most six.

Axial dipole zero block. After gauge and Noether reduction, the only surviving covectors are the three lifted rotations. The homogeneous zero block supplies the time and circle translations. Therefore

$$\text{coker } \mathcal{L}_{\text{smooth}} = \text{span}\{\zeta_H, \zeta_{P_x}, \zeta_{J_1}, \zeta_{J_2}, \zeta_{J_3}\}. \quad (6)$$

Covariant identification. The exact Taub bridge identifies $\langle \zeta_X, \mathcal{E}^{(2)}(u, u) \rangle$ with $\mu_X(u)$. This proves both necessity and sufficiency.

Remark 4.2. The theorem is stronger than checking a chosen set of modes. It includes arbitrary finite sums across the certified generic, exceptional, homogeneous, and twist blocks. It is weaker than an infinite-mode Sobolev theorem and weaker than causal solvability.

5 What changes in the bounded category

On $\mathcal{X}_{\text{bounded}}$, differentiation cannot invert a positive power of t , and a resonant exponential has no bounded primitive. The obstruction ledger consequently enlarges to

$$\{\mu_H, \mu_{P_x}, \mu_{J_i}\} \cup \{\mathcal{P}_{j,r}\} \cup \{\mathcal{R}_{j,a}\}. \quad (7)$$

The certificates contain fixtures in which the last two families are independent of all five moment maps. Hence the smooth cone is strictly larger than the bounded cone.

Theorem 5.1 (Standard generalized-zero bounded cone). *On the complete standard generalized-zero carrier with homogeneous coordinates (a, b, c, d, Q_e, W_x) and axial twist position/velocity (A, B) , bounded second-order extension holds exactly on*

$$a = b = Q_e = 0, \text{quad} B = 0, \text{quad} c, d, W_x \text{ arbitrary, quad} A \in \mathbb{R}^3 \text{ arbitrary}. \quad (8)$$

The surviving source has an explicit bounded correction.

The triangular positive-degree ideal supplies the proof. Its leading rows contain

$$\frac{15}{2}b^2t^2, \text{quad} \text{STF}(B \otimes B)t^2, \text{quad} Q_e a t.$$

They force $b = B = 0$ and $Q_e a = 0$. Intersecting with the five moment maps then forces $a = Q_e = 0$ on this carrier. The remaining homogeneous source vanishes, while constant twist position is tangent to an exact lifted mapping-torus family and has a constant polar correction.

Corollary 5.2 (Universal polynomial elimination). *For every point of the complete finite-support bounded cone,*

$$b = 0, \text{quad} B = 0, \text{quad} Q_e a = 0.$$

Oscillatory products cannot cancel these zero-frequency polynomial rows.

6 Natural transport columns

Several apparently dangerous global-oscillator products are derivatives of exact families and can therefore be classified without solving each harmonic source independently.

Proposition 6.1 (Electric and Wilson columns). *For every certified nonzero-frequency standard or extra primary, the electric-charge cross source has a bounded correction obtained by differentiating electromagnetic duality,*

$$F_\theta = \cos \theta F + \sin \theta \star_g F.$$

At the background,

$$h_{\text{cross}} = 0, \text{quad} f_{\text{cross}} = \star_g f + (D_g \star)[h] F_{\text{bg}}.$$

Its sphere period vanishes for $\ell \geq 1$, so it lifts globally on the fixed bundle. Every Wilson-oscillator source vanishes because $\delta A = W_x dx$ has $\delta F = 0$ and the local action depends on the connection through F .

This removes the complete Q_e and W_x oscillator columns from the bounded frontier. It does not remove the separate homogeneous condition $Q_e a = 0$.

Theorem 6.2 (Complete circumference column). *Let c vary the circle radius and let a certified oscillator have compact momentum k . The c -times-oscillator source admits a bounded correction if and only if*

$$c = 0 \quad \text{or} \quad k = 0.$$

For $k \neq 0$, the radius family gives a smooth secular correction but no bounded one.

Indeed, writing $R^2 = 1 + \eta c$,

$$\omega_R^2 = \frac{k^2}{R^2} + m_{\text{branch}}^2, \quad \partial_\eta \omega_R|_0 = -\frac{ck^2}{2\omega}.$$

Differentiating the exact family produces

$$\partial_\eta e^{-i\omega_R t}|_0 = \frac{ick^2}{2\omega} t e^{-i\omega t}.$$

The action-derived current is nonradical on every standard branch and extra multiplicity block, so the associated shell pairing cannot vanish when $ck \neq 0$. This is a genuine $\mathcal{R}_{j,a}$ obstruction, not a polynomial $\mathcal{P}_{j,r}$ obstruction.

7 Exceptional resonance as an independent obstruction

Moment-map cancellation does not guarantee bounded or even nonsecular second-order closure in a restricted exceptional carrier. For the axisymmetric exceptional $\ell = 1, k = 0$ axial and polar amplitudes (a_x, a_p) , a twist velocity can cancel the time-translation moment map. The two parity self-sources can also be tuned to cancel their even $L = 2$ resonance. But the required interior point has $a_x a_p \neq 0$, and the mixed axial $L = 2$ source has exact nonzero adjoint pairing

$$-\frac{8\sqrt{3}}{9}.$$

The boundary axes retain their individual self-obstructions. Thus the only axisymmetric two-polarization exceptional solution of the complete second-order equation is $a_x = a_p = 0$. The example proves that resonant cokernel equations contain information independent of global charges.

8 Current bounded frontier

After the certified eliminations above, the full bounded problem is sharply localized. The still-open rows are:

1. the homogeneous a -column against all oscillators;
2. the homogeneous d -column against nonzero-momentum oscillators;
3. shell resonances among bounded wave products and the remaining exceptional/global blocks;
and
4. the common zero locus of these functionals for arbitrary finite sums.

The electric, Wilson, and circumference columns are no longer open. The first full-time velocity fixture also sharpens the remaining d -column. Locality gives

$$\mathcal{E}^{(2)}[dt, u](t) = t \mathcal{E}^{(2)}[c, u](t) + \mathcal{E}^{(2)}[dt, u](0).$$

For the two axial extra representatives and the first polar representative, the coefficient of t vanishes. For the second polar representative it is the nonzero eight-row vector

$$(-36, 0, 164, -8\sqrt{3}i, 0, -100, -24, -20).$$

Thus the isolated fixture has polynomial zero locus $dz_{\text{polar},2} = 0$. The previously computed constant-shell adjoint matrix remains valid, but cannot by itself establish boundedness. Joint cancellation with the a -column and same-channel sources remains open. Any future completion must preserve the separation between positive-degree obstructions $\mathcal{P}$ and shell pairings $\mathcal{R}$.

Carrier or column	Smooth correction	Bounded correction
Complete finite-support cone	iff five moment maps vanish	additional $\mathcal{P}$ and $\mathcal{R}$ conditions
Standard generalized-zero block	classified	classified completely
Electric charge $\times$ oscillator	finite quasiperiodic	removable by duality transport
Wilson $\times$ oscillator	zero source	zero source
Circumference $\times k = 0$ mode	ordinary same-frequency	removable
Circumference $\times k \neq 0$ mode	secular radius derivative	obstructed
Velocity $d \times$ selected polar extra mode	smooth polynomial source	nonzero $\mathcal{P}_{j,1}$ unless canceled jointly
Exceptional axial-polar interior	secular criterion fails	obstructed
Remaining a, d oscillator columns	allowed by smooth theorem on moment cone	open

9 Interpretation

The solution set near a symmetric compact background is not generally a linear space. Its second-order approximation is a cone cut out first by a moment map and then, if uniform boundedness is demanded, by resonance conditions. Three points are important.

First, a linearly nonradical extra Weyl mode may fail to be tangent to the nonlinear fixed-charge solution space. This is nonlinear selection, not linear gauge removal. Second, mixing Einstein and extra directions can cancel the moment maps, so no conclusion follows from the sign of a pure branch alone. Third, even complete moment-map cancellation leaves bounded resonances. Smooth formal continuation and bounded perturbative stability are therefore different claims.

The result supplies the middle map in the programme's residual atlas:

$$X_\alpha \longmapsto \Omega(X_\alpha, \overline{X}_\alpha) \longmapsto \mu(X_\alpha) \longmapsto D^2\mathcal{E}[X_\alpha, X_\alpha].$$

Interaction sourcing, detector response, and BRST status are separate later maps.

10 What is not proved

The following remain outside the theorem:

- infinite harmonic sums and Sobolev completion;
- retarded or advanced solvability and nonlinear causal dependence;
- integration of a second-order jet to an exact or all-orders family;
- the final residual/background-stabilizer quotient;
- asymptotically flat, de Sitter, anti-de Sitter, or black-hole boundary conditions;
- observer coupling, particles, quantum BRST cohomology, and unitarity.

These exclusions are not cosmetic. In particular, Theorem 4.1 does not establish nonlinear stability.

11 Certificate and reproduction ledger

The manuscript is an OUTLINE-ALLOWED/DRAFT-ALLOWED integration of committed certificates. Its current source baseline is repository commit 35ae48835e4ef0bafb87bb94ff515446879d632b.

Claim	Authoritative certificate
Complete smooth finite-support cone	bridge/certificates/einstein_maxwell_weyl_complete_finite_harmonic_smooth_global_second_order.json
Standard generalized-zero bounded cone	bridge/certificates/einstein_maxwell_weyl_standard_global_bounded_second_order.json
Electric and Wilson transport	bridge/certificates/einstein_maxwell_weyl_electric_wilson_complete_oscillator_transport.json
Complete circumference column	bridge/certificates/einstein_maxwell_weyl_circumference_complete_oscillator_bounded_classification.json
First full-time velocity polynomial column	bridge/certificates/einstein_maxwell_weyl_d_ell2_extra_full_time_polynomial.json
Exceptional two-polarization resonance	bridge/certificates/einstein_maxwell_weyl_exceptional_ell1_two_polarization_resonance.json
Opposite-momentum smooth theorem	bridge/certificates/einstein_maxwell_weyl_opposite_momentum_smooth_global_second_order.json
Moment-map/Taub bridge	bridge/certificates/einstein_maxwell_weyl_moment_map_taub_bridge.json

Each certificate records its schema, imported hashes, exact commands, test tier, assumptions, and missing-object ledger. The independent verifier for each named certificate must pass before this manuscript advances to THEOREM-FROZEN. The current draft does not promote the still-open complete bounded cone.

12 Outlook

The immediate mathematical target is the full bounded finite-support cone. The immediate physical target is to compare its strata with the transferred Einstein/extra/Maxwell interaction tensor and with observer response. A later causal theorem would replace the finite secular right inverses by support-controlled Green operators or prove that this cannot be done.

The intended final synthesis is not that every fourth-order mode disappears. It is that each mode receives an exact disposition:

$$\boxed{\text{linear mode} \rightarrow \text{moment map} \rightarrow \text{resonance} \rightarrow \text{interaction and observation.}}$$

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